DEEP LEARNING IN FINANCE · COURSE PROJECT

Advanced Statistical Arbitrage with Reinforcement Learning

From discretized price-spread trading to sizing vega-neutral IV-spread pairs

Based on **Ning & Lee (2024)**, *Advanced Statistical Arbitrage with Reinforcement Learning*, arXiv:2403.12180 — extended to our own ATM vega pairs project.

rl-atm-vega-pairs

THE PAPER

Ning & Lee (2024): swap two hand-tuned pieces for two learned ones

Pairs trading has two phases — the paper replaces the traditional tool in each

PHASE	TRADITIONAL APPROACH	THIS PAPER
Part 1 · Which spread to trade	Hand-fit B (Distance Method or OU)	EMRT — B chosen to minimize empirical reversion time
Part 2 · When to trade it	Fixed threshold rule (e.g. trade at ±1σ)	Q-learning — a policy learned from experience

10 pairs
US sector pairs, formed 2022, traded 2023

2 benchmarks
Distance Method & OU trading

Daily Sharpe
headline metric, next slides

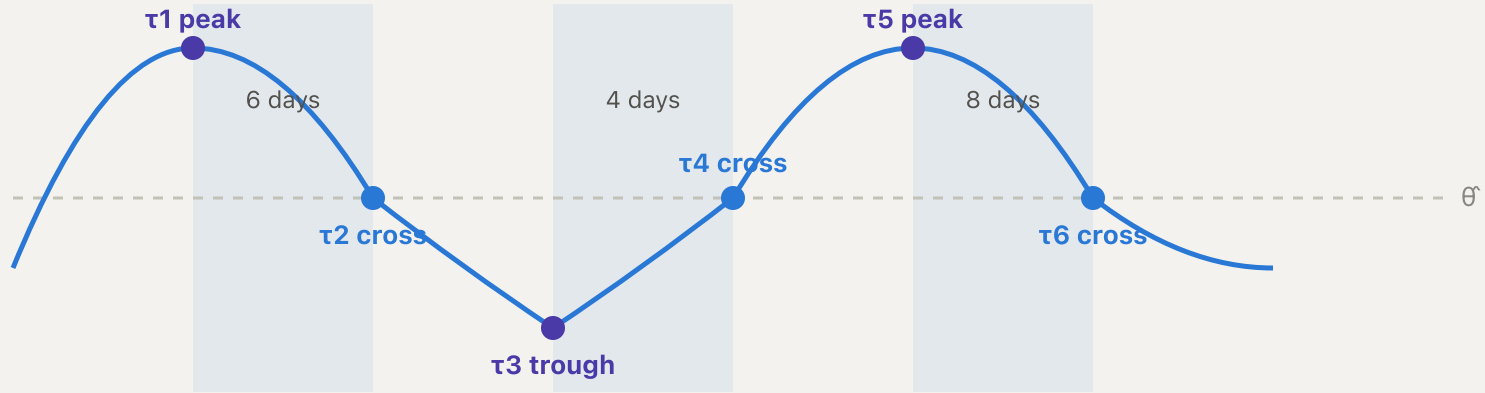
PART 1 · FORMATION

EMRT: average time from a big swing back to the mean

Find the meaningful peaks and troughs, time each one back to the mean line, average the times

$$X_t = S_{1,t} - B \cdot S_{2,t}$$

The spread. Each B gives a different spread; some revert quickly, some drift.



Shaded = stopwatch running (extreme → next mean crossing). A swing counts only if it moves ≥ C·s from the mean (C = 2, s = spread std dev). B is chosen to minimize this EMRT time.

PART 2 · TRADING

Trading as an MDP: state, action, reward

The agent trades the fixed spread $X_t = S_1 - B^*S_2$, one unit at a time

State S_t ↑ +1 | ↑↑ +2 | ↓ −1 | ↓↓ −2 → $S_t = [+1, +2, -1, -2]$

Last l = 4 daily % moves of the spread, each bucketed at k = 3% (small / big, up / down) → $4^4 = 256$ states

ACTION

- {sell −1, hold 0, buy +1}
- Flat → buy or hold; long → sell or hold; never short

$$R_{t+1} = A_t \cdot (\theta - X_t) - c \cdot |A_t|$$

Paid for trading **toward** the mean: buy below θ, sell above θ. Uses today's gap, not tomorrow's move.

- The state carries **no level information** — only the shape of recent moves.
- The agent is a **price-taker**: its trades don't move the spread.

PART 2 · TRADING

Q-learning: from one-step rewards to a 256 × 3 table

Each reward only looks at today; the future enters through the next state's Q-value

ONE STEP

R_{t+1}

Reward for today's action

→

DISCOUNTED TOTAL

$G_t = \Sigma_k \gamma^k R_{t+k+1}$

$= R_{t+1} + \gamma G_{t+1}$ (random)

→

EXPECTED TOTAL

$Q(s,a) = E[G_t \mid s, a]$

One fixed number per cell

$$Q(S_t,A_t) \leftarrow Q(S_t,A_t) + \alpha \cdot [\, R_{t+1} + \gamma \cdot \max_a Q(S_{t+1},a) - Q(S_t,A_t) \,]$$

The bracketed target $R + \gamma \cdot \max Q(S_{t+1})$ is a one-step stand-in for G_t (bootstrapping). Q converges; it doesn't accumulate.

$\alpha = 0.1$

learning rate

$\gamma = 0.99$

discount factor

$\varepsilon = 0.1 \rightarrow 0$

ε-greedy in training, greedy at test

PART 2 · ASIDE

Why the agent trains on simulated spreads

The reward needs something you never observe in real markets

PROBLEM

- The reward uses the **true** mean θ — unknown on real data
- One real price path is far too little data to fill 256 × 3 cells

FIX

- Simulate 10,000 one-year OU paths with known θ (μ=1, θ=1, σ=0.1)
- Train the Q-table there, then trade the **real** 2023 spread with it

- All headline results that follow are on **real** stock data; only training is synthetic.

RESULTS

RL wins on daily Sharpe in every pair

10 US sector pairs · B fit on 2022, traded in 2023 · daily closes from Yahoo Finance

Index	MSFT-GOOGL	CVS-JNJ	CL-KMB	V-MA	GE-BA
	DM Method				
DailyRet (%)	0.0446	0.0440	0.0659	-0.0244	0.2771
DailyStd (%)	0.4670	2.0692	1.2314	1.1796	3.9356
DailySR	0.0955	0.0213	0.0535	-0.0207	0.0704
MaxDD (%)	-2.1344	-24.6778	-13.6791	-10.7238	-18.1823
CumulPnL (%)	11.4443	5.7581	15.6385	-7.4888	64.2387
	OU Method				
DailyRet (%)	0.0327	-0.0073	0.0198	0.0342	0.0392
DailyStd (%)	0.4285	1.4950	0.7589	0.3475	0.3890
DailySR	0.0764	-0.0049	0.0261	0.0985	0.1007
MaxDD (%)	-2.1427	-25.6665	-5.7253	-1.0185	0.000
CumulPnL (%)	8.2443	-4.5179	4.298	8.7348	10.046
	RL Method				
DailyRet (%)	0.1344	0.0585	0.0826	0.0330	0.1679
DailyStd (%)	1.0754	0.7506	0.6000	0.3144	1.2803
DailySR	0.1250	0.0780	0.1377	0.1049	0.1312
MaxDD (%)	0.0000	0.0000	-1.9476	-0.6211	0.0000
CumulPnL (%)	37.7555	14.8895	22.2879	8.4248	48.8196

Table 3: Performance summary for trading mean reversion portfolios by baselines and the proposed RL method.

Index	OXY-XOM	WELL-VTR	PPG-SHW	VZ-TMUS	CSX-NSC
	DM Method				
DailyRet (%)	0.0373	0.0694	-0.0772	-0.112	0.0000
DailyStd (%)	2.0950	0.6555	1.0113	3.7311	0.0000
DailySR	0.0178	0.1058	-0.0764	-0.0300	0.0000
MaxDD (%)	-19.1535	-1.4004	-19.1196	-37.4779	0.0000
CumulPnL (%)	3.9194	18.2114	-18.5547	-36.1756	0.0000
	OU Method				
DailyRet (%)	0.0238	0.0539	0.0000	-0.0123	0.0199
DailyStd (%)	1.6012	0.5480	0.0000	1.3241	0.2879
DailySR	0.0149	0.0983	0.0000	-0.0093	0.0693
MaxDD (%)	-14.5001	-1.3798	0.0000	-18.652	0.0000
CumulPnL (%)	2.7812	13.9245	0.0000	-5.0869	4.9825
	RL Method				
DailyRet (%)	0.0609	0.0745	0.1124	0.0412	0.0496
DailyStd (%)	0.9446	0.6794	1.0600	0.8037	0.7101
DailySR	0.0861	0.1097	0.1061	0.0513	0.0698
MaxDD (%)	-2.0008	-3.2895	0.0000	-3.7306	0.0000
CumulPnL (%)	15.0791	19.6910	30.4559	9.9163	12.4263

Table 4: Performance summary for trading mean reversion portfolios by baselines and the proposed RL method.

- **Benchmarks:** Distance Method (B=1, trade at ±1σ) and OU trading (OU-fit B, trade at $\pm 0.5\sigma_{eq}$).
- Best daily Sharpe (DailySR) in **10 / 10** pairs. Mean daily Sharpe: DM **0.024**, OU **0.047**, RL **0.100**.
- Best or tied-best max drawdown in 9 / 10 pairs. Negative DailySR entries (e.g. V-MA, PPG-SHW under DM) are real — those benchmark strategies lost money on those pairs.

Tables 3–4, reproduced from Hong & Lee (2024). Caveat: one train/test split, hand-picked pairs, no significance tests; CSX-NSC RL win is 0.0698 vs. OU's 0.0693.